

# Project Design Phase

## Solution Architecture

| Field         | Details                                  |
|---------------|------------------------------------------|
| Date          | 18 June 2026                             |
| Team ID       | 24EG105H22                               |
| Project Name  | HouseHunt – House Rent Management System |
| Maximum Marks | 4 Marks                                  |

### Solution Architecture Overview

HouseHunt follows a 3-Tier Client-Server Architecture with a clear separation between Presentation, Business Logic, and Data layers. The API contract is defined in OpenAPI YAML and auto-generates type-safe React Query hooks and Zod validation schemas, ensuring consistency across frontend and backend.

#### Architecture Layers

| Layer                       | Component               | Technology                                             | Responsibility                                                                                         |
|-----------------------------|-------------------------|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------|
| Presentation Layer          | Web Browser (React App) | React 19, Vite, Tailwind CSS, shadcn/ui, Framer Motion | User interface for Owner, Renter, Admin; routing via Wouter; API calls via generated React Query hooks |
| API Gateway / Reverse Proxy | Replit Shared Proxy     | Path-based routing at /api and /                       | Routes /api/* to Express server; serves React app at /                                                 |
| Business Logic Layer        | REST API Server         | Node.js 24, Express 5, JWT (jsonwebtoken), bcryptjs    | Handles auth, property CRUD, booking workflow, admin approval; validates requests with Zod schemas     |
| Data Layer                  | Relational Database     | PostgreSQL 16, Drizzle ORM                             | Stores users, properties, bookings; enforces referential integrity                                     |
| Auth Layer                  | JWT Middleware          | jsonwebtoken, bcryptjs                                 | Issues/validates JWT tokens; role guard middleware (requireRole) enforces RBAC                         |
| Code Generation             | API Codegen             | Orval + OpenAPI spec (openapi.yaml)                    | Auto-generates React Query hooks and Zod schemas from OpenAPI contract                                 |

### Data Flow

| Step | Actor           | Action                                              | System Response                                                                                       |
|------|-----------------|-----------------------------------------------------|-------------------------------------------------------------------------------------------------------|
| 1    | User (any role) | Opens browser, navigates to HouseHunt URL           | React app loads; public property listings fetched from GET /api/properties                            |
| 2    | User            | Registers or logs in with email/password            | POST /api/auth/register or /login; bcrypt verifies password; JWT token issued; stored in localStorage |
| 3    | Owner           | Submits new property listing via dashboard form     | POST /api/properties with JWT; saved to DB with status='pending'                                      |
| 4    | Admin           | Reviews pending properties in Admin Panel           | GET /api/admin/properties; PATCH /api/admin/properties/:id /approve or /reject                        |
| 5    | Renter          | Searches properties, views details, submits booking | GET /api/properties with filters; POST /api/bookings with JWT                                         |
| 6    | Owner           | Reviews booking requests in Owner Dashboard         | GET /api/bookings/my; PATCH /api/bookings/:id/approve or /reject                                      |
| 7    | Renter          | Tracks booking status in Renter Dashboard           | GET /api/bookings/my; status shown as Pending/Approved/Rejected                                       |

## Component Diagram

| Component               | Sub-components                                                                                         | Interfaces                                                                            |
|-------------------------|--------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Frontend (React + Vite) | AuthContext, ProtectedRoute, HomePage, PropertyDetail, OwnerDashboard, RenterDashboard, AdminDashboard | Calls REST API via generated React Query hooks; reads JWT from localStorage           |
| API Server (Express 5)  | auth.ts routes, properties.ts routes, bookings.ts routes, admin.ts routes, auth middleware             | Exposes /api/* endpoints; validates with Zod; queries DB via Drizzle ORM              |
| Database (PostgreSQL)   | users table, properties table, bookings table                                                          | Accessed via Drizzle ORM; referential integrity via foreign keys                      |
| Shared Libraries        | lib/api-spec, lib/api-client-react, lib/api-zod, lib/db                                                | OpenAPI contract → codegen → hooks + schemas; DB schema shared via workspace packages |

## Security Architecture

| Security Concern     | Implementation                                                                                      |
|----------------------|-----------------------------------------------------------------------------------------------------|
| Authentication       | JWT tokens signed with SESSION_SECRET; stored in localStorage; sent as Authorization: Bearer header |
| Password Security    | bcryptjs with salt rounds=10; plain passwords never stored                                          |
| Authorization (RBAC) | requireRole('owner' 'renter' 'admin') middleware on protected routes                                |
| Input Validation     | Zod schemas validate all request bodies server-side (generated from OpenAPI spec)                   |
| CORS                 | Express CORS middleware configured for allowed origins                                              |
| Environment Secrets  | DATABASE_URL and SESSION_SECRET stored as Replit environment secrets; never hardcoded               |